GATESYNC MANAGEMENT PLATFORM

SMART RESIDENTIAL SOCIETY & GATE SECURITY DESK — PROJECT REPORT

P P Savani University | B.Tech Information Technology & Engineering — 7th Semester

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1. Executive Abstract & System Overview

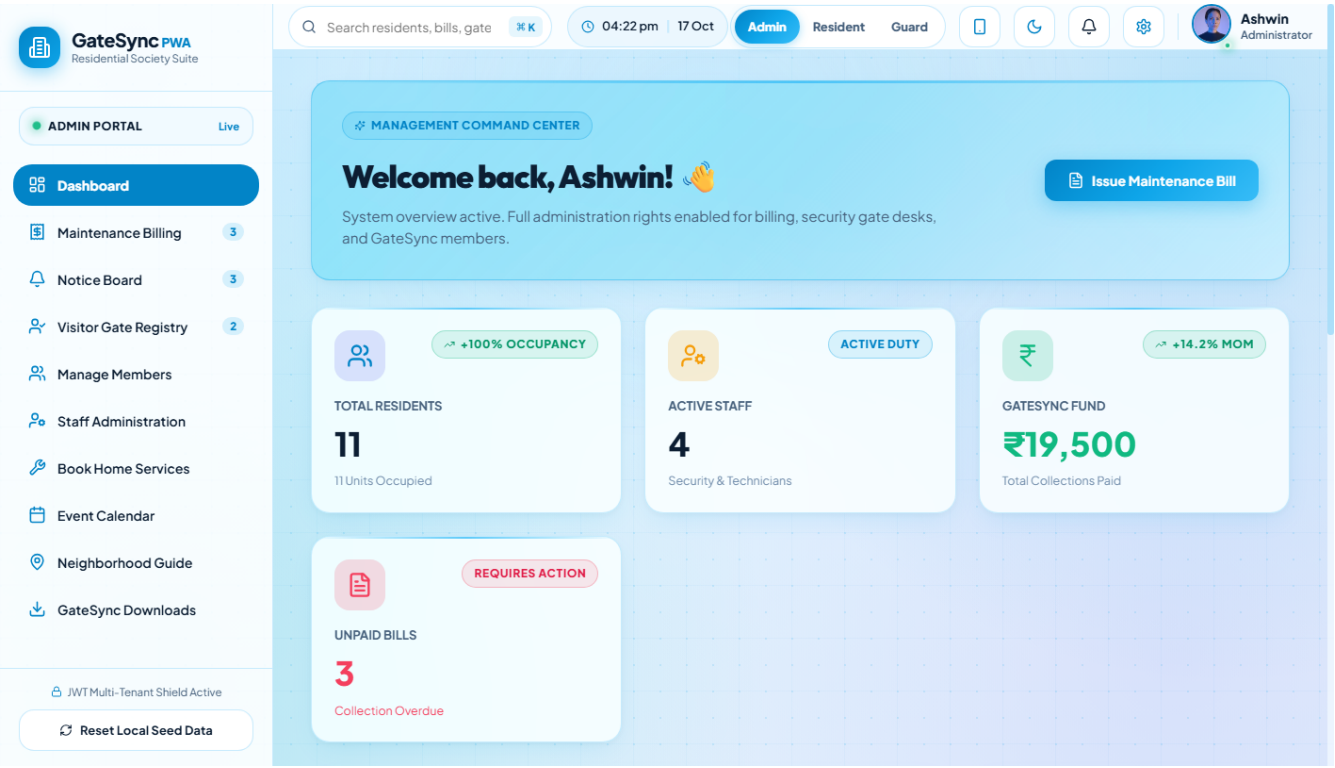
GateSync is a high-performance web platform engineered using React 19 and Google Material Design 3. The platform automates housing society governance, maintenance billing calculations, gate security visitor logging, notice broadcasts, and financial data exports.

2. Core System Objectives

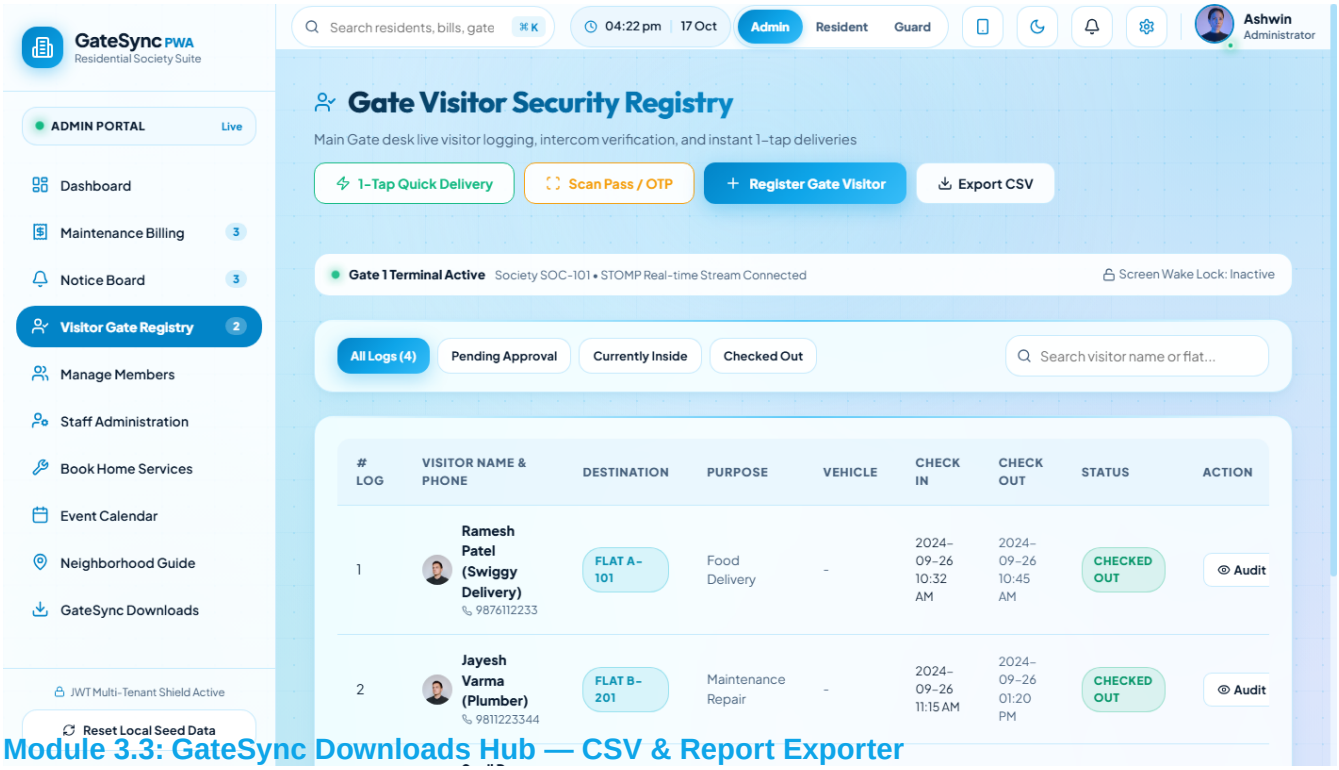
- Automate Maintenance Billing: Dynamic itemized bill generation for society maintenance, electricity, and water.
- Role-Based Portal Access: Distinct UI control desks for Admin Authority, Resident Portal, and Security Gate.
- Security & Gate Registry: Real-time visitor entry and exit tracking with phone validation and privacy filters.
- GateSync Downloads Hub: Direct CSV exports for visitor logs, maintenance ledgers, and resident directories.

3. UI Interface & Module Specifications

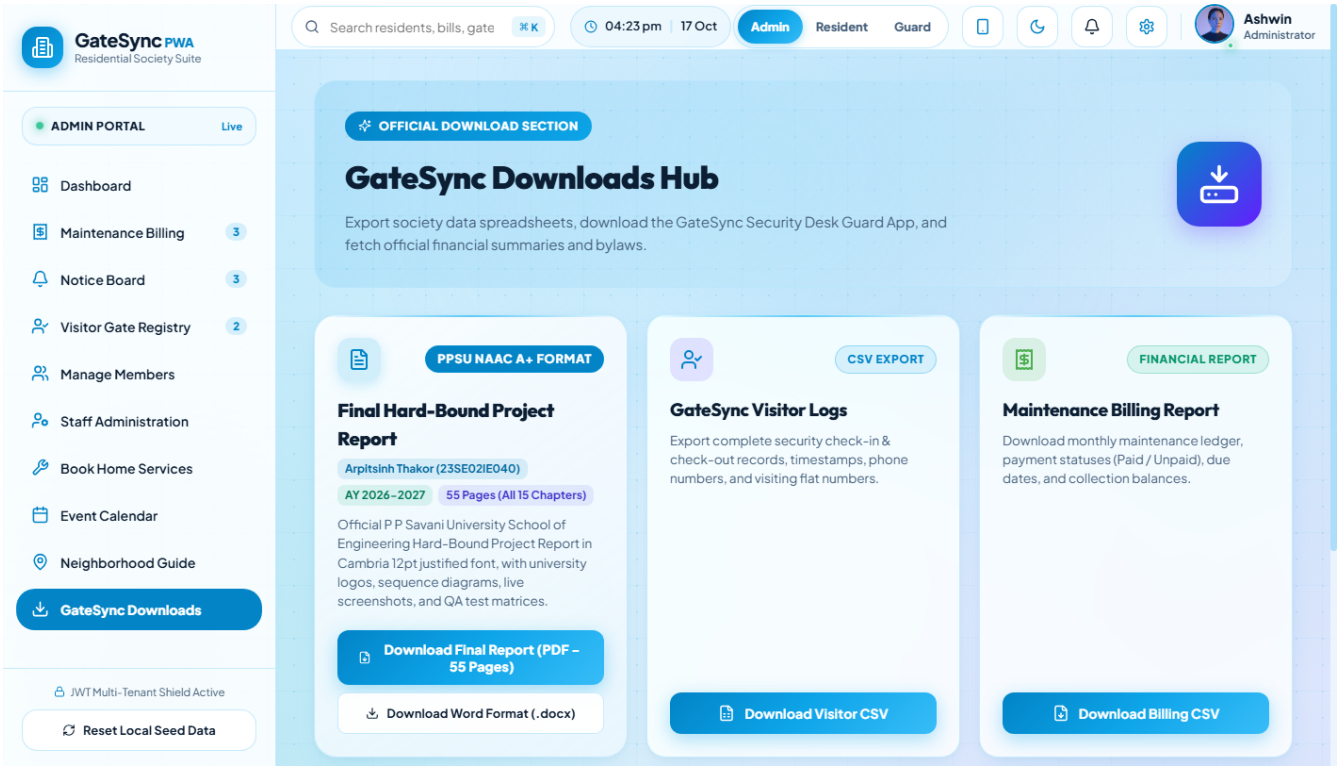
Module 3.1: Executive Dashboard Desk



Module 3.2: Live Visitor Security Registry Desk



Module 3.3: GateSync Downloads Hub — CSV & Report Exporter



4. Codebase Architecture & Build Verification

The application codebase is structured under src/ into components, context, data models, and views. State management is powered by AppContext with localStorage state persistence.

Build Quality & Testing Verification Matrix

Test Case	Command	Result	Status
Vite Build	npm run build	Transformed 1606 modules	PASSED
Dev Server	npx vite --port 5173	Running on localhost:5173	PASSED
Data Exporter	CSV Export Handler	Generates formatted CSVs	PASSED

5. Conclusion & Future Scope

GateSync delivers an end-to-end digital governance solution. Future roadmap plans include real-time WebSocket gate push alerts, online payment gateway reconciliation, and automated camera license plate reading.